$$-\frac{d}{dx}\left(K(x)\frac{du}{dx}\right) = 100x \qquad u(2)=0 \qquad \frac{du(0)}{dx} + u(0) = 20$$

$$K(x) = \begin{cases} 1 & \text{dla } x \in [0,1] \\ 2x & \text{dla } x \in (1,2] \end{cases}$$

$$\int_0^2 -\frac{d}{dx}\left(K(x)\frac{du}{dx}\right) v(x)\,dx = \int_0^2 100x\, v(x)\,dx$$

$$\int_0^2 -\frac{d}{dx}\left(K(x)\frac{du}{dx}\right) v(x)\,dx = \left[-K(x)\frac{du}{dx}v(x)\right]_0^2 + \int_0^2 K(x)\frac{du}{dx}\frac{dv}{dx}\,dx$$

dla $x=2$ $\quad u(2)=0 \Rightarrow v(2)=0$

dla $x=0$ $\quad \frac{du(0)}{dx} + u(0) = 20$

$$\Downarrow$$

$$-K(0)\frac{du}{dx}v(0) = -v(0)(20-u(0)) = -20\,v(0) + u(0)v(0)$$

$$\int_0^2 K(x)\frac{du}{dx}\frac{dv}{dx}\,dx = \int_0^1 \frac{du}{dx}\frac{dv}{dx}\,dx + \int_1^2 2x\frac{du}{dx}\frac{dv}{dx}\,dx$$

$$\int_0^1 u'(x)v'(x)\,dx + \int_1^2 2x\,u'(x)v'(x)\,dx - v(0)u(0) = \int_0^2 100x\,v(x)\,dx - 20\,v(0)$$

$$B(u,v) = \int_0^1 u'(x)v'(x)\,dx + \int_1^2 2x\,u'(x)v'(x)\,dx - v(0)u(0)$$

$$L(v) = \int_0^2 100x\,v(x)\,dx - 20\,v(0)$$